ANSWER KEY

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)REGULAREXAMINATION October 2026

Fifth SEMESTERSkill Enhancement Course (SEC)

MG5SECECO300 - SECURITY ANALYSISAND PORTFOLIO MANAGEMENT

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes Maximum Marks: 50

Remember(K),Understand(U),Apply(A),Analyse(An)Evaluate(E),Create(C),Skill(S) , Interest(I)and Appreciation(Ap)Students should attempt at least one question from each course outcome to enhance their overalloutcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries 1 marks

1. a) Initial Public Offering
2. b) Identifying investment objectives
3. a) TRUE
4. c) RBI

5a) Government expenditure and taxation

6. c) The security has the same systematic risk as the market
7. c) Economic analysis
8. a) A pool of assets
9. c) 50 securities
10. c) Establishing a long-term target allocation among different asset classes
11. b) Financial institutions, financial markets, financial instruments and financial services
12. c) Expected market return
13. c) A Premium
14. c) hammer
15. c) Coupon

Part B

Short Answer Type questions

Answer any 10 questions

Each question carries 2 marks

16. What is discounted cash flow analysis? State its basic principle.

Discounted Cash Flow (DCF) analysis is a valuation method used to estimate the value of an investment based on its expected future cash flows. The Time Value of Money (TVM), which states that a rupee today is worth more than a rupee tomorrow, so future cash flows must be discounted to their present value using a required rate of return.

17. Why is risk assessment important in security selection?

Risk assessment ensures that the chosen securities align with the investor's risk tolerance and financial goals. It helps in avoiding unexpected capital losses, enables proper diversification, and ensures an optimal risk-return trade-off in the portfolio.

18. Distinguish between coupon rate and market interest rate.

- Coupon Rate: The fixed annual interest rate paid by the bond issuer, calculated as a percentage of the bond's face (par) value. It remains constant throughout the bond's life. Market Interest Rate: The prevailing rate of return demanded by investors for similar bonds in the open market. It fluctuates based on economic conditions, inflation, and central bank policies.

19. What is a derivative instrument?

A derivative is a financial contract whose value is derived from the performance of an underlying entity, such as an asset (e.g., stocks, commodities), index, or interest rate. Common examples include futures, options, forwards, and swaps.

20. State the primary methods used to construct a passive index portfolio.

1. Full Replication: Buying all the securities in the index in the exact same proportions.

2. Stratified Sampling: Buying a representative subset of securities that match the index's key characteristics (e.g., sector weight, market cap).

3. Optimization: Using mathematical models to select a portfolio that minimizes tracking error relative to the index while managing transaction costs.

21. What is meant by the coupon rate of a bond?

The coupon rate is the annual interest rate paid by a bond's issuer to the bondholder, expressed as a percentage of the bond's face (par) value. For example, a 5% coupon rate on a ₹1,000 bond means ₹50 in annual interest.

22. What is meant by a financial benchmark?

A financial benchmark is a standard, index, or reference point (e.g., Nifty 50, Sensex, or a specific bond index) against which the performance, risk, and returns of a security, portfolio, or fund manager can be measured and evaluated.

23. Briefly explain the concept of industry analysis.

Industry analysis is the evaluation of a specific industry's economic, competitive, and market conditions to determine its growth potential, profitability, and risks. It often uses frameworks like Porter's Five Forces or the Industry Life Cycle to assess how macroeconomic factors and competitive dynamics impact companies within that sector.

24. What is the role of depositories in the structure of the Indian capital market?

Depositories (like NSDL and CDSL) hold securities (shares, bonds) in electronic (dematerialized) form. Their role is to enable seamless, secure, and rapid transfer of ownership, eliminate the risks of physical certificates (theft, forgery, damage), and reduce paperwork and transaction costs.

25. What is the relationship between Random Walk Model and Efficient Market Hypothesis?

The Random Walk Model suggests that stock price changes are independent, random, and unpredictable. This is a direct mathematical implication of the Efficient Market Hypothesis (EMH), which states that asset prices fully reflect all available information. Because new information arrives randomly, price adjustments must also be random, making it impossible to consistently "beat the market" through prediction.

26. Calculate the expected return.

Return 1 = 8% (Probability = 0.4), Return 2 = 18% (Probability = 0.6)

Expected Return =14%

27. What is an Investment Policy Statement (IPS)?

An Investment Policy Statement (IPS) is a formal, written document drafted between a portfolio manager and a client. It outlines the client's investment objectives (return goals), risk tolerance, investment constraints (liquidity, time horizon, taxes), and the guidelines/strategies the manager must follow when managing the portfolio.

28. How does the Single Index Model simplify portfolio analysis compared with the Markowitz model?

The Markowitz model requires calculating the covariance between every possible pair of assets, which is computationally massive for large portfolios. The Single Index Model simplifies this by assuming that all securities are correlated only through their common relationship with a single market index. This reduces the required estimates to just expected return, beta, and residual variance for each asset, plus the market's expected return and variance, making it highly practical**.(liberal valuation)**

Part C

Short Essay Type questions

Answer any 3 questions

Each question carries 5 marks

29. Explain the cyclical indicator approach to security analysis.

The cyclical indicator approach uses macroeconomic data to forecast the phases of the business cycle (expansion, peak, contraction, trough) and adjust investment strategies accordingly. Indicators are classified into three types:

1. Leading Indicators
2. Coincident Indicators:
3. Lagging Indicators:

30. What is Markowitz Portfolio Theory? Harry Markowitz's Modern Portfolio Theory (MPT), introduced in 1952, revolutionized investment management by focusing on the portfolio as a whole rather than individual assets.

Key principles:

1. Risk-Return Trade-off
2. Risk Measurement
3. Diversification:
4. Efficient Frontier

31. Explain the major monetary variables that influence the security market.

Monetary variables, controlled primarily by the central bank (RBI), directly impact liquidity and valuation in the security market:

1. Interest Rates (Repo/Reverse Repo
2. Money Supply
3. Inflation

4. Reserve Ratios (CRR/SLR): Lowering the Cash Reserve Ratio (CRR) or Statutory Liquidity Ratio (SLR) frees up bank funds for lending, increasing market liquidity and supporting higher asset valuations.

32. Analyse the importance of trading indicators and explain its functions.

Trading (technical) indicators are mathematical calculations based on historical price, volume, or open interest, used to forecast future price movements.

They remove emotional bias from trading, provide objective entry/exit signals, and help confirm trends or potential reversals.

1. Trend Identification
2. Momentum Measurement
3. Volatility Assessment
4. Overbought/Oversold Signaling

33. Explain the Arbitrage Pricing Theory (APT) and describe the role of multiple systematic risk factors in determining security returns.

Developed by Stephen Ross, the Arbitrage Pricing Theory (APT) is an alternative to the CAPM. It posits that the expected return of a financial asset can be modeled as a linear function of various macroeconomic factors or theoretical market indices.

Role of Multiple Systematic Risk Factors:

APT recognizes that multiple macroeconomic forces simultaneously impact asset returns. Common factors include:

1. Changes in inflation rates.
2. Changes in industrial production (GDP growth).
3. Changes in interest rate structures (yield curve shifts).
4. Changes in credit risk spreads.

Each factor carries its own risk premium. A security's total expected return is determined by its unique sensitivities (factor betas) to each of these macroeconomic risks. This makes APT more flexible and realistic than CAPM, as it can be customized to reflect the specific economic drivers relevant to a particular asset or market.

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